Sparse Retrieval with Learned Term Weights

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Abstract

We study sparse retrieval models that learn a weight for every term of a document. Our model keeps the inverted index of classic search engines while matching the effectiveness of dense retrievers on standard benchmarks.

1 Introduction

Information retrieval systems answer a query with a ranked list of documents. Classic systems score documents with term statistics such as BM25, which needs no training data and runs fast on an inverted index. Neural retrievers instead encode queries and documents into dense vectors and compare them with a dot product.

Dense retrievers need a nearest neighbour index and a great deal of memory. Sparse retrievers with learned weights keep the inverted index and its efficiency, and close most of the gap in effectiveness. This paper describes such a model and the conditions under which its improvements hold, together with an analysis of its failure modes on long and heterogeneous collections.

2 Method

Each term t of a document d receives a weight from a contextual encoder. The score of a document for a query q is the sum of the weights of the terms they share:

$$s(q, d) = \sum_{t \in q \cap d} w_t(d) \cdot w_t(q). \tag{1}$$

We train the encoder with a contrastive loss over one positive document and a set of negatives:

$$\mathcal{L} = -\log \frac{\exp s(q, d^{+})}{\sum_{d'} \exp s(q, d')}. \tag{2}$$

A regulariser keeps the representations sparse, so that the index stays small and queries stay fast.

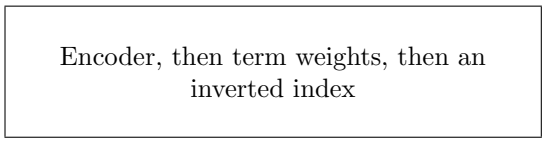


Figure 1: The model: an encoder gives each term a weight, and the weights go into an inverted index.

3 Experiments

We evaluate on three collections of news, scientific abstracts and web pages. Our model improves the mean reciprocal rank over BM25 by twelve points and matches a dense retriever of the same size, while answering queries four times faster on a single processor. Effectiveness drops on very long documents, where a few frequent terms dominate the weights; splitting documents into passages recovers most of the loss.

4 Conclusion

Learned term weights make sparse retrieval competitive with dense retrieval at a fraction of its cost. Future work will study multilingual collections and the interpretability of the weights the encoder assigns.